# C-Series Compressors: Maintenance and Troubleshooting

## 1. Scope and safety

This guide covers preventive maintenance, fault finding and the oil change procedure for C-Series compressors. Set points and limits are given in C-OM-06.

Before any maintenance: stop the compressor with the stop button, isolate it according to SOP-LOTO-001, close the outlet isolation valve and release all pressure from the separator vessel according to SOP-PRS-002. The oil and the airend can be hotter than 80 °C directly after stopping.

## 2. Service intervals

| Interval | Task |
|---|---|
| Daily | Check oil level, condensate drain and controller messages. |
| Every 500 hours | Clean the air filter pre-filter mat. Check the cooler for dirt. |
| Every 2,000 hours | Replace the air filter element. Clean the cooler fins. |
| Every 4,000 hours or 12 months | Change the oil and the oil filter. Replace the air filter element. |
| Every 8,000 hours | Replace the oil separator element. Check the minimum pressure valve. |
| Every 8,000 hours | Check the thermostatic valve and the condensate drain. |
| Every 24,000 hours | Airend bearing inspection by authorised service. |

Replace the oil separator element earlier if the separator differential pressure exceeds **0.8 bar** (controller message A:SEP), or if oil carry-over into the network is observed.

### *2.1 Oil specification*

Use only the synthetic compressor oil specified for C-Series compressors (ISO VG 46, 4,000-hour oil). Do not mix with mineral oil. Oil capacity: C-22: 14 L, C-37: 22 L, C-55: 30 L.

## 3. Fault table

| Symptom | Probable cause | Remedy |
|---|---|---|
| High discharge temperature | Low oil level | Top up oil; look for leaks. |
| High discharge temperature | Dirty or blocked cooler | Clean the cooler fins with compressed air from the fan side. |
| High discharge temperature | Hot air recirculation / poor ventilation | Check room ventilation and ducting. |
| High discharge temperature | Faulty thermostatic valve | Replace the thermostatic valve element. |
| High discharge temperature | Clogged oil filter | Replace the oil filter. |
| High discharge temperature | Ambient above 40 °C | Improve ventilation. |
| Delivery pressure too high, safety valve blows | Unloading pressure set too high | Check controller set points. |
| Delivery pressure too high | Inlet valve does not close | Check inlet valve and solenoid valve. |
| Delivery pressure too low | Air demand higher than capacity | Check for network leaks; add capacity. |
| Delivery pressure too low | Clogged air filter | Replace the air filter element. |
| Oil in compressed air | Separator element damaged or saturated | Replace the separator element. |
| Oil in compressed air | Scavenge line blocked | Clean the scavenge line and its nozzle. |
| High vibration | Loose mounting or worn coupling element | Check anti-vibration mounts and coupling. |
| High vibration | Airend bearing wear | Contact authorised service. |
| Motor overload trip | High pressure, low voltage, airend damage | Check set points and supply voltage; turn the airend by hand. |

## 4. Troubleshooting high discharge temperature

Use this procedure for warning A:EDT (105 °C) or shutdown S:EDT (110 °C).

1 **Read the event log** on the controller. Note the temperature, the ambient temperature and the loaded hours at the event.

2 **Check the room.** Measure the ambient temperature near the cooling air inlet. Look for recirculation of hot cooler air and for blocked ventilation openings.

3 **Check the oil level** with the compressor stopped and depressurised. Low oil level is the most frequent cause of high temperature.

4 **Inspect the cooler.** Remove the cooler cover and inspect the fins. Clean with compressed air (maximum 6 bar) blowing from the fan side, or with a soft brush. Do not use high-pressure water.

5 **Check the oil filter and thermostatic valve.** If the cooler outlet stays cold while the airend is hot, the thermostatic valve is stuck in bypass. Replace the element.

6 **Check the oil.** Dark or sludgy oil has lost its cooling capacity. Change the oil and the oil filter (section 5).

7 **Restart and monitor.** After the repair, run the compressor loaded for at least 30 minutes and confirm the discharge temperature is between 80 and 95 °C.

Do not bypass or raise the 110 °C shutdown. At higher temperatures the oil can oxidise and there is a risk of fire in the separator vessel.

## 5. Oil and oil filter change

1 Run the compressor until the oil is warm (about 60 °C); warm oil drains faster and carries contaminants out.

2 Stop the compressor with the stop button. Isolate and lock out according to SOP-LOTO-001.

3 Close the outlet isolation valve and release the pressure according to SOP-PRS-002. Check that the separator vessel pressure gauge reads **0 bar**.

4 Slowly open the oil filler plug one turn to release any remaining pressure.

5 Drain the oil from the drain valve of the separator vessel and the oil cooler into a container. Dispose of used oil as hazardous waste.

6 Replace the oil filter. Lubricate the new filter gasket with oil and tighten by hand, three quarters of a turn after contact.

7 Close the drain valves and fill with new oil to the maximum mark of the sight glass.

8 Refit the filler plug, remove the locks and start the compressor. Run for a few minutes, stop, depressurise and check the oil level again.

9 Reset the service counter on the controller and record the oil change in the maintenance ticket.